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# Mechanical Mechanisms of Mesophyll Development in *Arabidopsis thaliana*

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Abstract here

into here?

## 1. Introduction

Ideal flow:

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– Intro sentance(?)
– Second sentance giving more context to why morphogenesis may be interesting beyond plant physiologists.
– Two sentance description of leaf physiology.
– Central Question
– Narrow to Arabidopsis as a first (and representative) step.

The structure of plant leaves is optimized both for photosynthesis and for mechanical stability. For the plant physiologist understanding how leaf tissue develops will shed light on how evolution can tune phenotype through structure. For biophysicists and materials engineers understanding leaf development will help understand naturally occurring multifunctional materials, and help design novel bioinspired materials. Most leaves are composed of four main tissues that form distinct layers, the top and bottom layers are epidermal tissue often one or two cells in depth with pores (stomata) that allow gas to enter the interior of the leaf. Within the leaf are two layers of mesophyll, the upper inner layer is palisade mesophyll, vertically elongated approximately cylindrical cells, and the lower inner layer is spongy mesophyll. As its name suggests spongy mesophyll is highly porous, which allows gas percolate through the tissue facilitating the exchange of $CO_2$ and $O_2$ required for photosynthesis. Structurally, spongy mesophyll is the most striking of these tissues with its lobed cells that expose a large proportion of cell surface area to the air. The mechanics of spongy mesophyll development in leaves offers a unique question within plant morphogenesis: How does this tissue generate the complex geometries required to create porosity, and the interfaces required to carry out its photosynthetic functions while remaining structurally sound?

Despite the importance of leaf tissue development to feeding the world, leaves' role in terrestrial plant survival, and the possible photosynthetic bioinspired technologies that might be derived from plant leaves, tissue development in leaves remains poorly understood.

Physiological studies of leaf development and mesophyll tissue indicate that a combination of local mechanics and local rules primarily govern the mechanics of spongy mesophyll development rather than a genetically prescribed blueprint for leaf physiology. Selective cell wall strengthening [**<empty citation>**], cell wall stretching [**<empty citation>**], and de-adhesion [**<empty citation>**] have each been observed during spongy mesophyll development and each have been hypothesized to be required to generate mature structures. However, while there has been excellent work describing the biochemical machinery critical to these features [**<empty citation>**], to the authors knowlege there have been no studies modeling the combined, and relative, mechanical roles of each of these features during spongy mesophyll morphogenesis. To date, we are aware of only two studies modeling the cell-level microstructure during leaf development, one study of the epidermis, and another study of spongy meosphyll, however both of these studies were conducted in 2D, meaning key feature inherently 3D features like bands of thick material that encircle lobes, but also restrict growth in the out of plane direction could not be directly modeled.

Here we present simulations that explicitly capture the 3D cell shape changes and validate them with experiential images of Arabidopsis thaliana.

What I want to say in the rest of the intro:

- What is poorly understood?
- ie where is the knowlege gap this this helps with?
- a more detailed description of the approach.

The mechanics of spongy mesophyll development in leaves offers a unique question within plant morphogenesis: How does this tissue generate the complex geometries required to generate porosity, and the interfaces required to carry out its photosynthetic functions while remaining structurally sound? Complex leaf mesophyll cell shapes have been noted as far back as the early 1900s [**haberlandt_physiological_1914**] and in recent decades large strides have been made identifying physical and biochemical evidence for the different phenomena that come together in the morphogenesis of spongy mesophyll [**borsuk_structural_2022**, **paternis_morphogensis_2005**]. In many species, including *Arabidopsis thaliana*, spongy mesophyll consists of cells with lobes forming disordered porous networks which start out as dense polyhedral cells but over the course of development become highly porous [**borsuk_structural_2022**, **zhang_live_2021**] (a few more citations here perhaps). This geometry is apparently helpful for gas exchange, but presents a puzzle in how air spaces are formed, creating holes in the tissue, without destroying the tissue. Additionally cells take on highly irregular, lobed, shapes with 'arms' reaching out towards one another [**borsuk_structural_2022**]. While significant progress has been made identifying mechanisms for porosity generation, and cell shape change, to date the mechanics have yet to be combined *mechanics* into a single mechanical model of mesophyll tissue growth.

At the organ level there has been a great deal of modeling of leaf growth, with several models put forward that can well describe the curvature and interesting boundaries of leaves [**zhao_microtubule-mediated_2020**, **boudon_computational_2015**], but these models either explicitly assume continuum material or do not allow for air space to be created. Similarly at the molecular and cell-signaling levels much has been accomplished, identifying the molecular machinery responsible for pressure control within cells [**beauzamy_flower_2014**, **morgan_osmoregulation_1984**,

**monshausen_force_2013**], cell wall production [**cosgrove_building_2022**], and controlled expansion [**cosgrove_disentangling_2**] as well as adhesive forces [**daher_how_2015**]. In between tissue, and molecule, the cell and multi-cellular mechanics of morphogenesis are less well understood, to our knowledge there have only been two published studies proposing mechanistic, cell-level, developmental models of leaf tissue growth, both of those carried out in 2D [**sapala_why_2018**, **treado_localized_2022**].

Part of the difficulty in modeling mesophyll morphogenesis is that, as in many tissues, there is thought to be a combination of mechanical and biochemical feedback and control. For example genetic controls are known to impact the overall shapes of leaves through local patterning, drastically altering local growth rates to go between small serrations or full leaflets [**kierzkowski_growth-based_2019**], while a combination of simple elaso-plastic polymers combined with cell signaling is thought to control cell wall growth and expansion [**cosgrove_building_2022**, **zhang_live_2021**, **treado_localized_2022**, **paternis_microtubule_1992**].

While genetic control and cell patterning is no doubt important to overall leaf shape, and the sequencing of growth stages including the end of cell division which occurs while the leaf is relatively small, the microstructure of mesophyll, particularly, mesophyll in the bulk of more simple leaves appears to be primarily determined by local forces and physical constraints [**zhang_live_2021**, **treado_localized_2022**]. We hypothesize that microstructure is primarily determined by turgor pressure, adhesive bond breaking, and inhomogeneous cell wall expansion controlled by cell contact geometry.

We present a first-of-its-kind model for mesophyll morphogenesis that brings together the hypothesized mechanisms of development into a high resolution 3D computational model and validated with structural measurements from experimental data. Using a both confocal microscopy and microcomputed tomogrophy (microCT) scans we quantify cell and tissue structure in mesophyll across development to validate the computational model. We measure porosity of the tissue ($\mathcal{P}$), the ratio of cell-cell contact surface area to total cell surface area ($A_{cc}/A_c$), cell asphericity ($\mathcal{A}$), and cell elongation ($\mathcal{E}$), to robustly quantify compare the experimental and computational development trajectories.

We take into account four key features: turgor pressure, the selective deposition and expansion of cell wall material according to cell geometry, cell-cell adhesion, and selective elongation along the growth along the Basal-Apical direction. We quantify the adhesive strength relative to turgor pressure required to create pore space. Finally we demonstrate which physical parameters structure is most sensitive to and highlight the capability of this model to produce varied microstructure qualitatively similar to the variation across microstructure in different species.

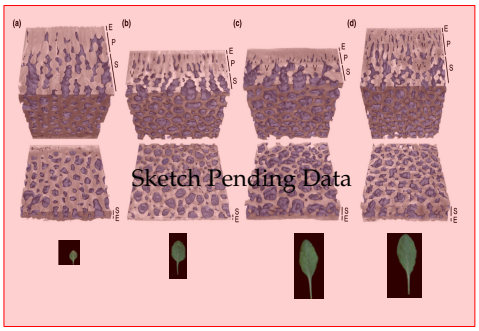



Figure 1: Images of developing arabidopsis leaves at different stages of development. (a)-(d) 3D confocal microscopy images of an arabidopsis leafs at progressively later stages in development. Air space is false colored blue, with cell material in gray. The volumes have been cut into two and split open to show cross sections of spongy mesophyll. The different layers of tissue have also been labeled along the right side of each frame as follows epidermis, upper and lower, E, palisade mesophyll, P, and spongy mesophyll, S. Panel (e) shows photos of the leaves that were images to create panels (a)-(d) in the same order.

## 2. Methods

Leaf growth begins with an apical meristem budding off the stem of plant, which grows into a leaflet [**lv_overview_2023**, **zhang_research_2022**], while doing so growing into different cell types with four distinct layers: the upper epidermis, the lower epidermis, and between them the palisade mesophyll on the upper internal layer and the spongy mesophyll on the lower internal layer. (See Fig. 1 (a)-(d).) As it's name suggests the spongy mesophyll structure is porous. Like the other tissues the spongy mesophyll begins as a confluent tissue, then through a transformation of cell shape and connectivity it generates the porosity we visible in Fig. 1(a)-(d). It is widely understood that the driving force for volumetric expansion for cells is the turgor pressure within the cells pushing out. To begin this transformation the confluent tissue begins to disadhere where three (or more) cells come together forming intercellular (IS) spaces [**panteris_microtubule_1993**, **apostolakos_microtubules_1991**]. This creates the first Cell-Air (CA) surfaces on the spongy mesophyll cells. Over the

course of the development these regions grow through a combination of further disadhesion (including full breaking of contact) and expansion of the cell walls.

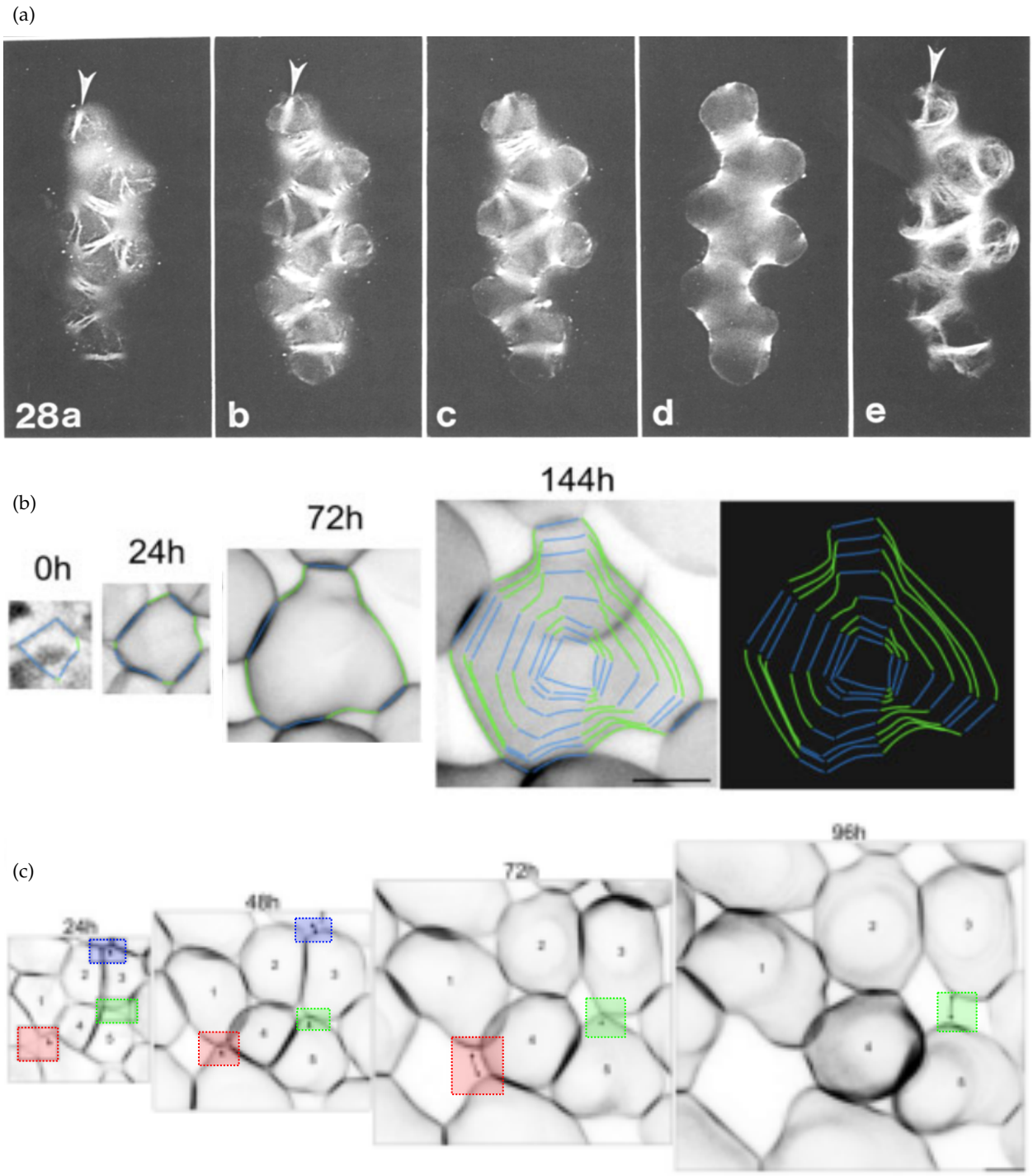



Figure 2: Biophysical mechanisms of mesophyll growth. (a) Cell wall 'banding' in *Adiantum capillus veneris* ([**panteris_microtubule_1993**]). Immunofluorescence (a-d) and Caleofluor (e) images of a mesophyll cell from a *Adiantum capillus veneris* leaflet. Panels (a)-(d) show the locations of microtubule bundles (bright white) in different focal planes which correlate highly with the regions of thicker cell wall (e). The differential cell wall growth in a cell wall across development is shown in (b). A close up of an *Arabidopsis Thaliana* mesophyll cell is shown across a 144h time span. At eight times (four shown) the cell wall outline was traced with cell-cell interfaces traced in blue, and cell-air interfaces traced in green. All outlines are shown overlaying the cell at time 144h, and, for clarity shown without the cell in the background. From these image one can observe that the cell-cell interfaces grow very little or not at all, and that far more growth occurs in the cell-air interface. This image is adapted from Zhang et al. [**zhang_live_2020**]. (c) Cell-cell adhesion and breaking. A group of *Arabidopsis Thaliana* mesophyll cells in early development grow and in this process three cell-cell contacts are broken. The breaks are annotated with arrows and highlighted in red, blue, and green across the image sequence. This image is also adapted from Zhang et al. [**zhang_live_2020**].

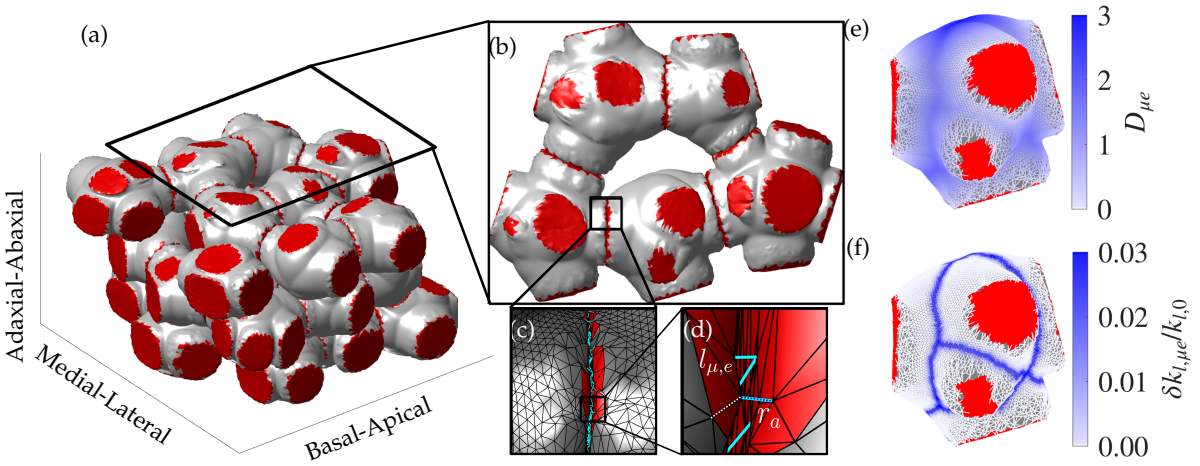



Figure 3: Mesophyll tissue model. (a) A snapshot of a growth simulation, 27 cells with periodic boundary conditions, cell surfaces exposed to air are shown in light gray, while cell surfaces contacting other cells, including periodic copies are red. (b) Top Down view of five cells in the top layer of the cells in (a). (c) A cell-cell interface, the triangles of the mesh are now added to show the resolution of the model. The cells are made up these triangles and the forces on the cells are calculated at the vertices. (d) Close up of the mesh and interface. Geometric parameters in the model are shown: edgelength $l_{\mu,e}$, the white dotted line and $r_a$ the distance between two adhered vertices is shown in dotted blue and cyan. Other adhesive bonds are shown in cyan. (e) The normalized distance to contact $D_{\mu e}$ is shown for the cell in the lower center of (b). As in (a)-(d) cell-cell contacts are shown in red, the rest of the surface shows the distance to the nearest contact. (f) The change in cell wall modulus $\delta k_l/k_{l,0}$ for a timestep is visualized for the same cell as in (e).

We believe that, while the underlying biochemical processes are complex, there are four key physical processes governing cell growth and shape change during spongy mesophyll morphogenesis. First growth is driven by the internal pressure in the cells. Second, cell-cell interfaces are adhered to one another via breakable adhesive bonds that break as the cells grow apart (See figure X.X). Third, cells lay down 'bands' of cell wall material in hoops (?) that circumscribe the cell and create regions of differential extensibility of the cell wall (See figure X.X). While the precise mechanism of these bands is not known (Adam?), it appears that the bands, as well as the microtubules associated with them are more present in regions far from cell-cell contacts. Fourth, and related to the idea of differential extensibility, the cell walls expand in regions of the cell wall not touching other cells (along the IS) (See figure X.X).

## (a) Computational Model

To better understand mesophyll growth, test ideas regarding the biomechics of growth, and examine the available designs in our proposed design space, we have developed a computational model for mesophyll growth that recapitulates changes in cells, and tissue, as mesophyll grows and develops due to the key features described above.

We represent the cell wall of each spongy mesophyll cell as a triangular mesh and model the mechanics of the cells using the vertices, edges, and faces of these meshes. The each cell, $\mu$, has a shape potential, $U_\mu^{cell}$, representing it's turgor pressure and cell wall strength. In addition there are potential energies representing cell-cell adhesive and repulsive contacts, $U^{ad}$ and $U^{rep}$ respectively. Finally, the system itself has an 'extended' potential, $U^{boundary}$, that maintains a fixed pressure at the boundary. All together the potential energy that determines the forces within system is:

$$U = \sum_{\mu=1}^{N_c} U_\mu^{cell} + U^{ad} + U^{rep} + U^{boundary}, \tag{2.1}$$

where $N_c$ is the total number of cells in the system. Each cell's potential energy has two terms one for the internal pressure and the other for the cell wall strength:

$$U_\mu^{cell} = -P_{turgor} V_\mu + \sum_{e=1}^{N_{\mu E}} \frac{k_{l,\mu e}}{2} \left( l_{\mu e} \breve{} l_{\mu e,0} \right)^2, \tag{2.2}$$

where $P_{turgor}$ is the turgor pressure on the inside of the cell wall, and $V_\mu$ is the volume of cell $\mu$. The summation term represents the potential energy of the cell wall where each edge, $e$, in the mesh of cell $\mu$ is treated as a harmonic spring, with spring stiffness $k_{l,\mu e}$, current length $l_{\mu e}$, and rest length $l_{\mu e,0}$. $N_{\mu E}$ is the number of edges in cell $\mu$.

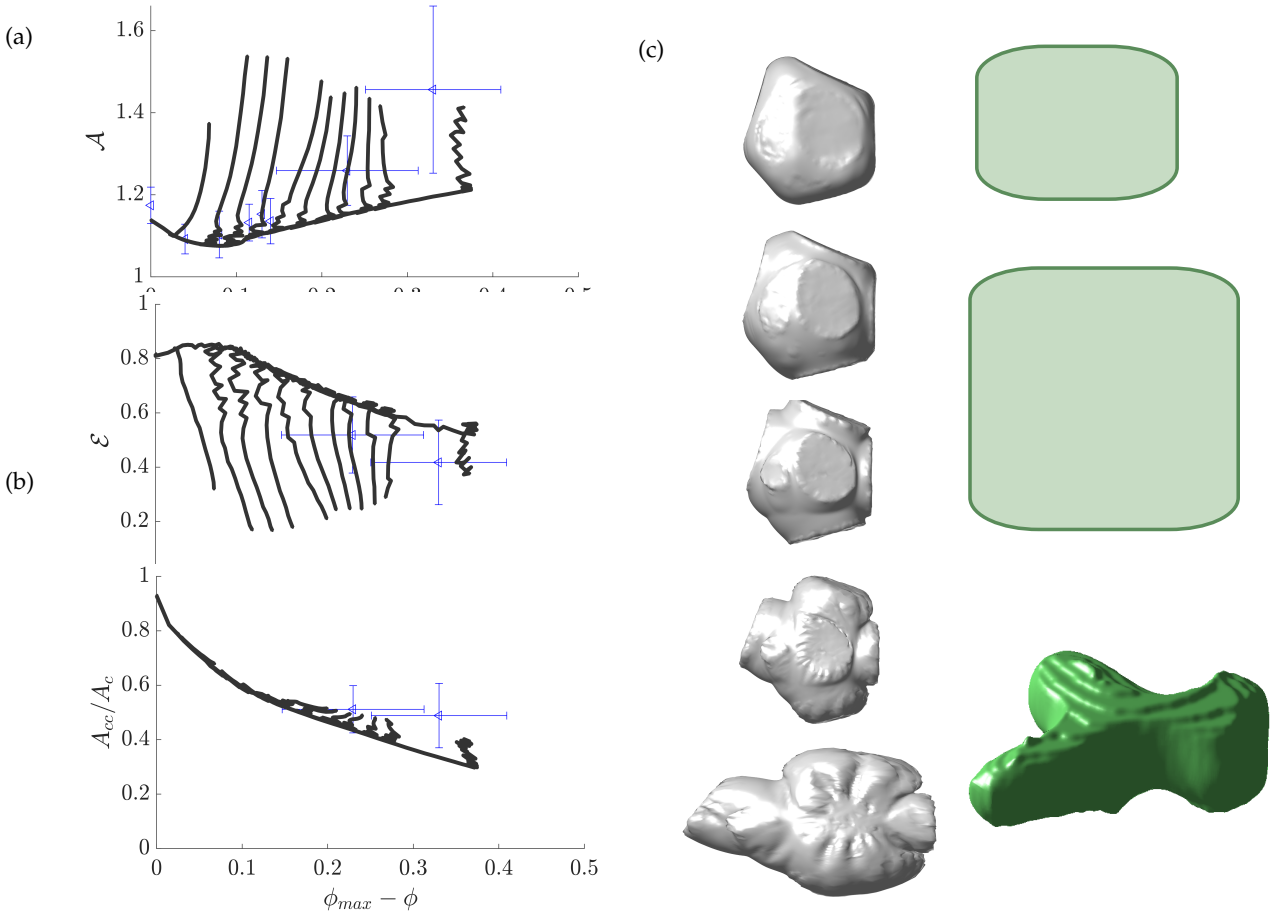



Figure 4: Growth model compared to experiment. (a) Asphericity, mean $\mathcal{A}$ versus volumetric expansion. The starting point of this expansion is taken to be the mean volume for data set with the smallest mean volume for experiment, and the mean volume of the simulated cells when they are confluent. Measurements from experiment are shown in colors with error bars indicating the standard deviation of the values. The data in black are results from the simulation. (b) Mean local porosity, $\phi^{max} - \phi$, vs normalized volume. Again measurements from experiment are shown in color with error bars indicating the standard deviation measured values, while simulation results are shown in black. (c) Images of cells at progressive stages of development, with simulation data on the left and experiment on the right.

The primary way cells interact with one another is through adhesive contacts that bond neighboring cells. We represent this bonding also through spring-like potentials, in the following potential energy:

$$U^{ad} = \sum_{a=1}^{N_{ad}} \frac{k_{ad}}{2} \left(r_a - r_{a,0}\right)^2, \tag{2.3}$$

where $N_{ad}$ is the number of adhesive bonds in the system, $k_{ad}$ is the stiffness of the springs representing these bonds, $r_a$ is the current length of the bond representing the $a$th adhesively bonded pair of vertices, and $r_{a,0}$ is the rest length of that bond.

In addition, a purely repulsive potential prevents cells from overlapping with one another:

$$U^{rep} = \sum_{\mu=1}^{N_c} \sum_{v=1}^{N_{\mu V}} \sum_{\lambda=1}^{N_c} \sum_{f=1}^{N_{\lambda F}} \frac{k_{rep}}{2} \left(\sigma - d_{\mu v,\gamma f}\right)^2 \times \Theta\left(\sigma - d_{\mu v,\lambda f}\right), \tag{2.4}$$

where $N_{\mu V}$ is the number of vertices on cell $\mu$, $N_{\lambda,F}$ is the number of faces on cell $\lambda$, $k_{rep}$ is the stiffness of the repulsive interactions, $\sigma$ is the range of the repulsive interactions, $d_{\mu v,\lambda f}$ is the shortest distance between vertex $v$ on cell $\mu$ and face $f$ on cell $\lambda$, and $\Theta$ is the Heaviside step function which ensures that pairs of vertices and faces father apart than the interaction range contribute no potential energy.

Finally we have the boundary potential energy that maintains a set pressure, $P_{boundary}$, on the the system:

$$U^{boundary} = P_{boundary} V_{box}, \tag{2.5}$$

where $V_{box}$ is the size of the periodic box. For more details on the boundary conditions see Appendix **??**.

## (b) Growth Procedure

We apply the growth protocol to this system mirroring the mechanisms described above.

We initialize a confluent system with average asphericity $\mathcal{A} = 1.17$, and packing fraction $\phi \approx 0.975$ which we define as $\phi^{max}$, and pressure $P_{boundary} = 0$. We set each rest length in the cell wall $l_{\mu e,0}$ to the current length, $l_{\mu e}$, and adhere vertices on different cells that interact via the purely repulsive potential, $U^{rep}$.

Starting from confluence growth is simulated by a series of growth steps. In each growth step the system is energy is minimized using F.I.R.E. V2 **[guenole_assessment_2020]**. Next, the graph distance $d^{ad}_{\mu e}$ from each edge to the nearest adhered vertex is calculated (see figure). From these measurements we also calculate the graph distance of each edge to the nearest ridge vertex $d^{ridge}_{\mu e}$. Ridge vertices are defined to lie equidistant to two or more cell-cell contacts (up to discritization of the mesh). (See Fig. 3(b).)

Using $d^{ridge}_{\mu e}$ we define a $\Gamma_{k_{l,\mu e}} = \left(1 - \exp(-d^{ridge}_{\mu e}/\sigma_{kc})\right)^{-1}$ which decays from one to zero as distnace increases from the ridge. It specifies where we will increase $k_{l,\mu e}$ (see Fig. 3e). This is meant to mimic the constricting bands laid down by the cell. At each growth step the edge stiffnesses increase as,

$$k_{l,\mu e} \leftarrow k_{l,\mu e} + k_{l,\mu e,0}\delta k_l \exp(-D_{peak,\mu e}/\sigma_{k_l}), \tag{2.6}$$

where $k_{l,\mu e,0}$ is the value $k_{l,\mu e}$ is initialized with, and $k_{l,\mu e,0}\delta k_l$ is the maximal rate of stiffness growth.

Similarly, using $d^{ad}_{\mu e}$ we define a $\Gamma_{l_{\mu e,0}} = \exp(-d^{ad}_{\mu e}/\sigma_{kc})$??? which specifies a region close to the ridge where we will increase $k_{l,\mu e}$ (see Fig. 3d).

To model the breaking adhesive bonds, all bonds under strain greater than $E_{ad}$, $\frac{l_{\mu e} - l_{\mu e,0}}{l_{\mu e,0}} > E_{ad}$, are broken and these bonds are removed.

## (c) Measurements From Experiments

# 3. Results

# 4. Conclusions

# A. Appendix 1

# Outline

(i) Introduction

(a) Mesophyll tissue is a a structurally interesting material. It must be optimized to conform to several, sometimes competing, functions.

(b) more about structure – what is known what is not known. it is interesting to think that there is some combination of fateing/global ordering/local emergence that leads to the overall structures and shapes we see in plants.

(c) Our pictures of growth are somewhat developped at the molecular level, as well as the continous organ level. However there is a knowlege gap in the modeling of cell-level development, particularly mesophyll tissue where there is strong evidence for particular mechanisms but to date no models that bring these together with the physics to validate and predict 'what if something were changed'.

(d) in this paper we do just that, we bring together leading ideas in mesophyll devleopment together with physical simulations to bring about a first of its kind tissue model of leaf mesophyll.

(e) we examine each underlying mechanical thing and bring out what is necessary to recapitulate the strcutres we oveerve in experimental systems using these mechanics in silico.

(f) finally we make predictions for what might happen if each of these mechanics were to be 'knocked out' and identify the mechanics most sensitive in the structure.

(ii) Methods
  (a) Mechanisms
  (b) Computational Model
  (c) Perhaps how we measure the metrics?
(iii) Results
(iv) Agreement with Wildtype
(v) Relationship of Parameters with Phenotype
(vi) Conclusion(s)

# B. Appendix 2

## (a) Full Description of Procedures

(i) A system of $N_c$ DPs is initialized with $k_c = 1, k_a = 10, k_v = 10, k_l = 10^{-5}, k_b = 0, P_0 = 0$. They are placed in a fixed aspect ratio box with pressure $P_{set}^{initial} = 10^{-3}$. Each cell has the same number of vertices $N_v = 10242$, and the cells are arranged in three layers of equal volume, though the cells within them are poly disperse.
(ii) Initially the three layers are populated using a soft sphere packing simulation.
(iii) Then, while keeping cells to thier layers with harmonic walls that divide the simulation box into three layers of equal size (in z), the system is brought to confluence. With a target packing fraction $\phi_{target} = 0.99$ and $\mathcal{A}_{target} = 1.17$
  (a) First the system is brought up to pressure using nose hoover NPH dynamics (to $P_{set}^{initial}$)
  (b) Next to achieve the target mean $\mathcal{A}$ and $\phi$, the volumes of the cells are iteratively shrunk.
  (c) This process terminates once $\phi > \phi_{target}$ & $\mathcal{A} > \mathcal{A}_{target}$ or one has been achieved and progress is not being made in the other.
(iv) At this point the layer-separating walls are removed.
(v) (reduce vertex size doesn't happen as it does not trigger)
(vi) Next the surface mesh becomes a spring surface mesh: The new parameters are: $k_v = 0, k_a = 0, k_{l,\mu e} = 1, k_b = 0, k_c = 1$. Additionally, $l_{0,\mu e} = l_{\mu e}$, The pressures in each cell are set $P_\mu = 0.01$
(vii) Next vertices in (or within 1 percent of) contact are adhered to one another with $r_{a,0} = r_a$ and $k_{ad} = 1$
(viii) To mimic the hypothesized checmically driven breaking of bonds, bonds are removed from each cell until they have on average `target_mean_contact_number=4` in-layer cell-cell bonds (while not allowing fewer than 3 in-plane bonds), out of plane bonds are not modified.
(ix) For the growth proceedure the z direction of the box is fixed while the x and y are free to independently grow and shirnk to maintain $P_{set} = 0$ pressure.
(x) After these changes fire minimization is run and the growth starts.
(xi) growth proceedure
Pressure growth step:
  (a) (adhesvie bond strengthening) if $\phi < \phi_{transition} = 0.96$, use $E_{ad} = 0.0077$ AND don't update cell walls, otherwise $E_{ad} = 0.03$
  (b) update cell walls:
    i. Exposed walls have plasticity $\beta_{exposed} =$ `minimum_plasticity` and contact walls have plasticity $\beta_{contact} =$ `plasticity_parameter`
    $l_{0,\mu e} = l_{0,\mu e} + \beta_{exposed|contact}(l_{0,\mu e} - l_{\mu e})$
    ii. $k_l$s are updated to create bands. The band locations are local maxima on the distance map to contacts. A small region around that is grown at a rate $\delta k_l$, the region is set by $\sigma_{kl}$
    iii. Any adhesive bonds that are streched more than $E_{ad} = 0.0077$, for $\phi > 0.96$ or $E_{ad} = 0.03$ otherwise, are broken. $k_{ad}$ rescaled cell by cell such that $N_{\mu,ad} k_{ad}$ remains constant ad $N_{\mu,ad}$ remains constant.

Ethics. Insert ethics text here.

Data Accessibility. Insert data access text here.

Authors' Contributions. Insert author contribute text here.

Competing Interests. Insert competing text here.

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